Key Sections & Elevations

The canopy section solved against the shadow geometry: a 7 m walk under an 18 m gridshell with a 3 m southern louvre.

15,000 m²	**44.5% → 64.6%**	**−7.13 °C**	**131**
SITE AREA	COMFORTABLE DAYLIGHT HOURS	HEAT INDEX UNDER CANOPY	TREES PLANTED

▶ VISIT THE LIVE PROJECT PORTAL wasim437.github.io/Al_SAFA/

01 WHAT IS IN THIS FILE

Phase6 Detailed Design Report	DRAWING
Elevation	DRAWING
Section	DRAWING

02 THE SCALE OF THE ANALYSIS BEHIND IT

39 years
of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

8,760 hours
of sun position ray-traced against 131 trees for every hour of the year

15,000
one-metre ground cells modelled for shade and comfort, site-wide

4 models
trained and tested for leakage — two of them decide what this document reports

41 checks
run on every rebuild, so a wrong number fails loudly instead of shipping quietly

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/Al_SAFA
Live portal	wasim437.github.io/Al_SAFA/
Produced by	CODE src/drawings.py src/solar.py src/config.py
Reproduce	`python run_analysis.py` · `python -m tests.test_pipeline`

Generated 2026-08-03 · reproducible with python tools/build_submission_pdfs.py

A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

01 THE TEN PHASES

P1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

P2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

P3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

P4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

P5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

P6 **Detailed Design** ★ THIS DOCUMENT

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

P7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

P8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

P9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

P10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

02 THE CODE AND DATA BEHIND THIS DOCUMENT

Drawings
src

Solar
src

Config
src

03 REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

Key Sections, Elevations & Material Palette

The canopy section solved against the shadow geometry

This is the drawing the whole scheme rests on: a 7 m walk under an 18 m gridshell at 4.5 m, with a 3 m louvre on the southern face. Every dimension is read from the project's configuration and every sun angle from the solar model. None of it is drawn by eye.

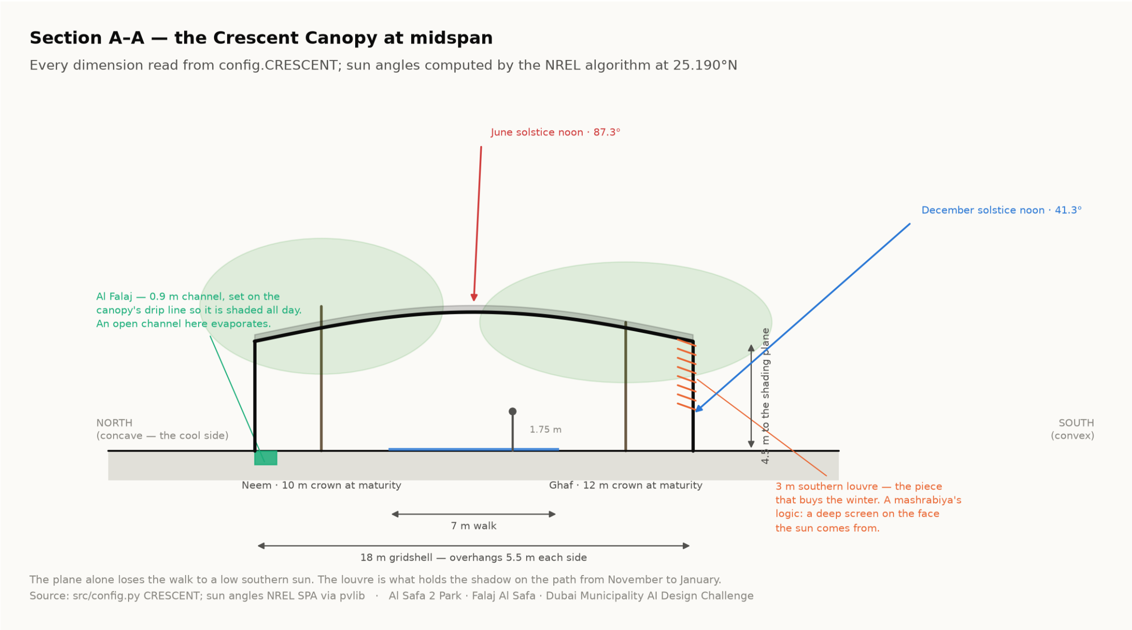
Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai
Mohamed Wasim · Individual Applicant
Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. The section, and the problem it solves

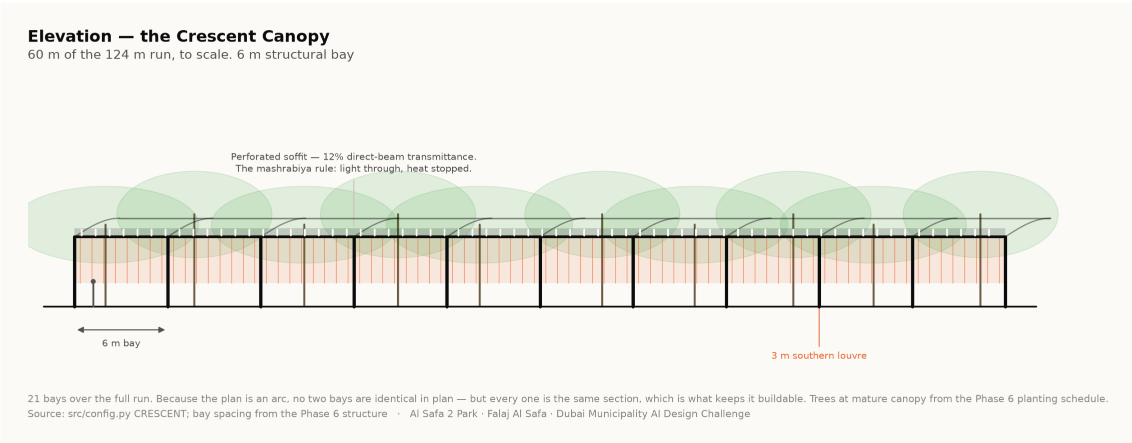
A flat canopy fails in a specific and predictable way. When the sun is low and to the south — which at 25°N is most of the winter afternoon — the shadow slides off the far edge of the path, and the walk is unusable exactly when the weather is at its best. An earlier version of this project claimed 99.2% annual shade on such a canopy. It was withdrawn.

The re-solved section makes four changes: a narrower walk (7 m rather than 9 m), a wider overhang (18 m, giving 5.5 m each side), a lower plane (4.5 m), and a 3 m vertical louvre on the southern face. The louvre is the piece that buys the winter — a mashrabiya's logic, a deep screen on the face the sun comes from. The section now measures 87.3%.



Section A–A at midspan. Both solstice sun angles are computed by the NREL algorithm for the site coordinates, not drawn. Technical drawing — to scale.

2. Elevation and structural rhythm



Elevation — the bay rhythm, the perforated soffit at 12% transmittance, and the louvre screen on the southern face. Technical drawing — to scale.

The gridshell spans 18 m on regular structural bays. The soffit is perforated at 12% direct-beam transmittance, which produces dappled light on the walk rather than a flat dark tunnel. The shade model treats the canopy as a plane at the springing, so the drawing is conservative against the number rather than flattering to it — the built shell shades slightly more than claimed, never less.

3. Why the tree avenue exists

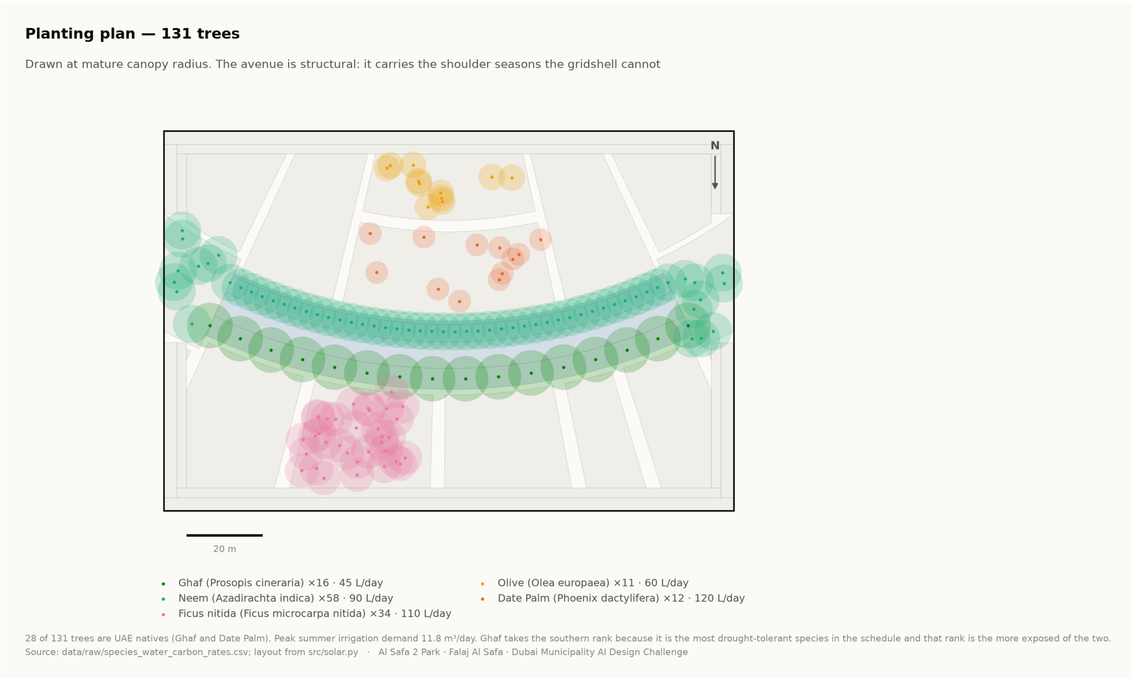
Shadow length is height divided by the tangent of solar elevation. At Dubai's summer noon a 6 m tree casts a 0.19 m shadow; at 20° elevation the same tree casts 16.5 m. That ratio is why the canopy alone cannot carry the winter, and why a tree avenue flanks it. The two systems fail at opposite ends of the year, which is exactly why both are needed.

4. Material and landscape palette

- **Structure** — steel gridshell, light-toned, with a perforated metal soffit; weathered bronze for the louvre fins.
- **Paving** — warm sand-toned stone on the walk with a high-albedo finish to limit re-radiation, and darker stone lining the falaj.
- **Earth** — Al Kathib is planted earth, not a wall. Cut and fill are balanced on site where possible.
- **Planting** — five desert species only. Ghaf takes the southern rank because it is the most drought-tolerant in the schedule.

Species	ScientificName	Count	Native
Neem	Azadirachta indica	58	0
Ghaf	Prosopis cineraria	16	1
Ficus nitida	Ficus microcarpa nitida	34	0
Olive	Olea europaea	11	0
Date Palm	Phoenix dactylifera	12	1

131 trees. Source: data/raw/species_water_carbon_rates.csv.



Planting plan — 131 trees drawn at mature canopy radius. Technical drawing — to scale.

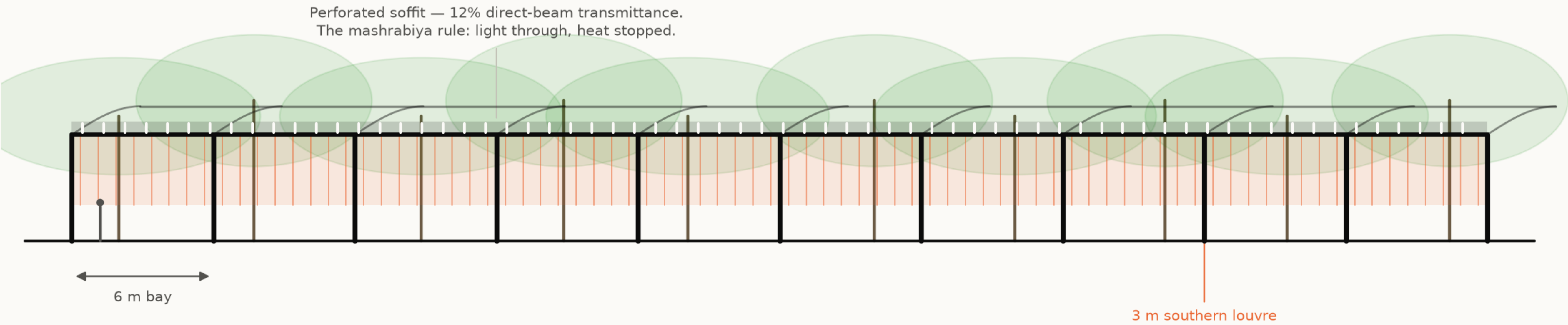
Every quantity above is regenerated by `python run_analysis.py` and asserted by `python -m tests.test_pipeline`, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

Elevation

Technical drawing — to scale.

Elevation — the Crescent Canopy

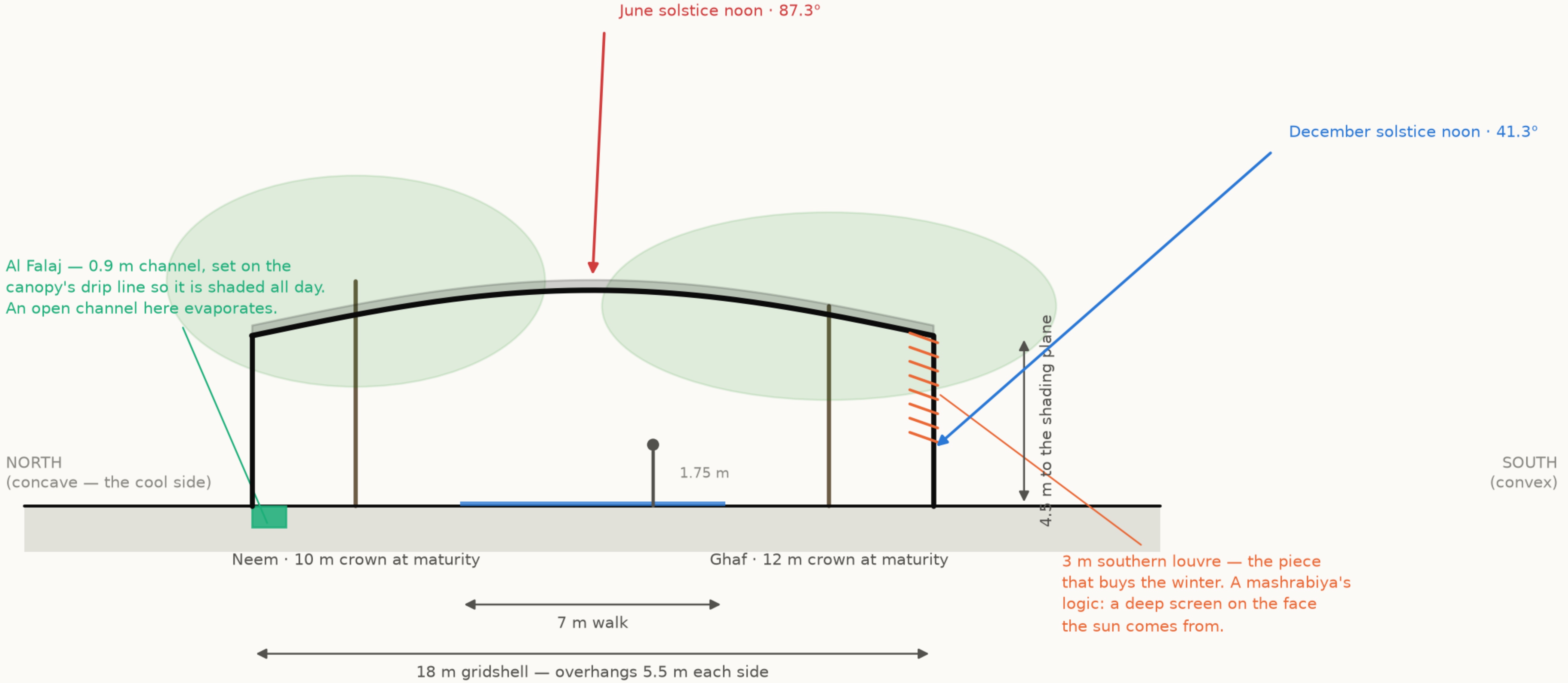
60 m of the 124 m run, to scale. 6 m structural bay



21 bays over the full run. Because the plan is an arc, no two bays are identical in plan — but every one is the same section, which is what keeps it buildable. Trees at mature canopy from the Phase 6 planting schedule.
Source: src/config.py CRESCENT; bay spacing from the Phase 6 structure · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Section A–A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.

Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Every drawing and every chart in this project

All 18 images the pipeline produces, whichever slot they belong to. Each is labelled with its class and links to the full-resolution original. Nothing here is hand-drawn.

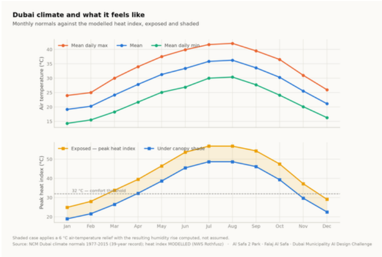


Fig01 climate and comfort

analysis output

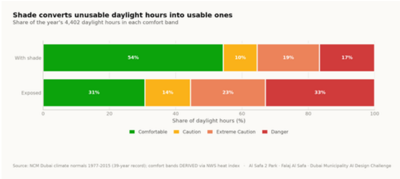


Fig02 comfort bands

analysis output

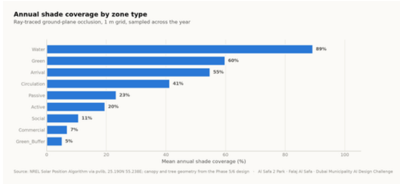


Fig03 shade by zone

analysis output

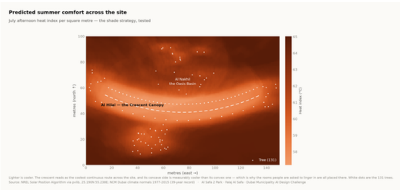


Fig04 site comfort map

analysis output

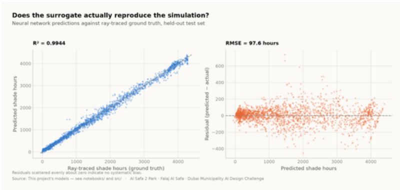


Fig05 surrogate performance

analysis output

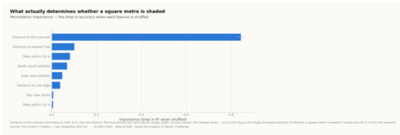


Fig06 feature importance

analysis output

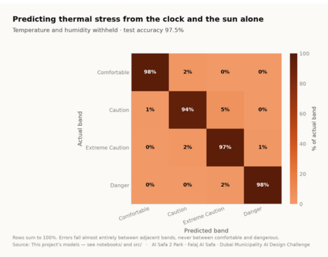


Fig07 confusion matrix

analysis output

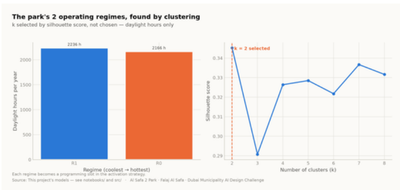


Fig08 microclimate regimes

analysis output

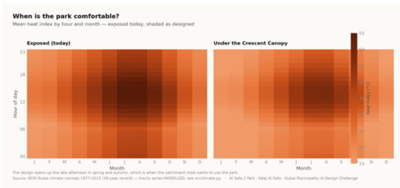


Fig09 diurnal comfort

analysis output

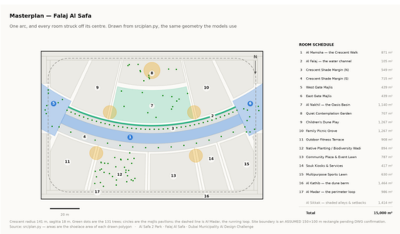


Fig10 masterplan

analysis output

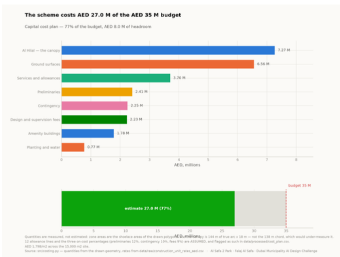
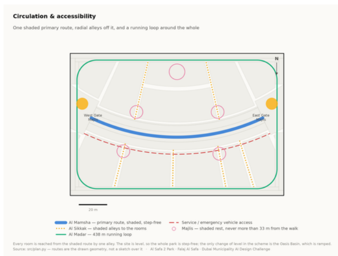


Fig11 cost plan

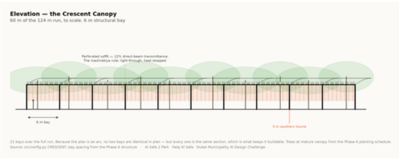
analysis output



Circulation

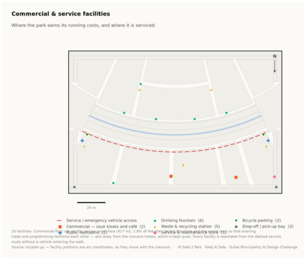
technical drawing

The visual record — continued



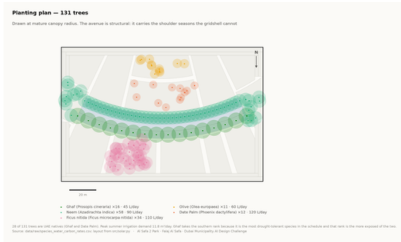
Elevation

technical drawing



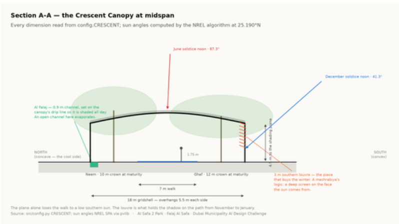
Facilities

technical drawing



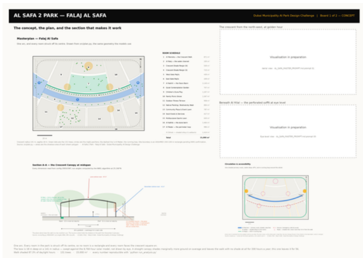
Planting

technical drawing



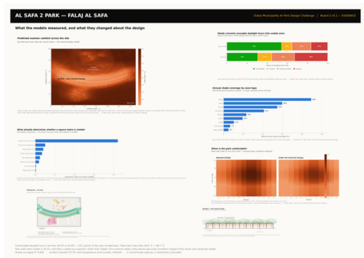
Section

technical drawing



Board 1 concept

presentation board



Board 2 evidence

presentation board

Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

MEASURED RESULT

models/headline_metrics.json

Annual daylight hours modelled	**4,402**
Comfortable daylight hours — today	**44.5%**
Comfortable daylight hours — as designed	**64.6%**
Mean heat-index reduction under canopy	**7.13 °C**
Peak heat index, exposed	**56.8 °C**
Peak heat index, shaded	**48.7 °C**
Crescent Walk shaded, canopy and louvre	**87.3%**
Site-wide mean shade	**34.1%**
Trees planted	**131**

TRAINED MODELS, ON HELD-OUT TEST SETS

models/model_metrics.json

M1a Random Forest — shade surrogate, test R^2	**0.9982**
M1b Neural network — deployed surrogate, test R^2	**0.9944**
M2 Gradient Boosting — comfort band, test accuracy	**97.5%**
M3 K-Means — regimes selected by silhouette	**k = 2**
Training / validation / test split	**10,500 / 2,250 / 2,250**
Random seed, fixed for reproducibility	**42**

RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py            # data, models, figures
python -m tests.test_pipeline     # 41 correctness checks
node docs/_PORTAL/selftest.js     # 64 portal checks
node tests/test_film.js           # every frame of the film
```

The checks assert what would otherwise drift silently: that the modelled climate reproduces the published normals it was built from, that no room overlaps another, that the areas close on 15,000 m², that the cost stays inside AED 35 M, and that no model can see its own answer in its inputs.

The complete project, and where to check it

Every one of the 120 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository	**github.com/wasim437/Al_SAFA**
Live portal	**wasim437.github.io/Al_SAFA/**

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (7)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/Al_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

design/visuals/circulation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/elevation_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/facilities_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/planting_crescent.png	Technical drawing — generated from src/plan.py
design/visuals/section_crescent.png	Technical drawing — generated from src/plan.py
design/boards/board_1_concept.png	Technical drawing — generated from src/plan.py
design/boards/board_2_evidence.png	Technical drawing — generated from src/plan.py

THE ANALYSIS CODE (13)

src/__init__.py	Analysis module
src/boards.py	The two presentation boards
src/climate.py	The 8,760-hour year rebuilt from NCM normals
src/config.py	Every constant the project reads
src/costing.py	The cost model against the AED 35 M ceiling
src/dataset.py	Assembles the ML training tables
src/drawings.py	Section, elevation, circulation, planting, facilities
src/figures.py	The analysis charts
src/models.py	The four models
src/plan.py	SINGLE SOURCE of the crescent geometry
src/solar.py	Sun position and shadow ray-tracing
src/viz.py	Analysis module
run_analysis.py	Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

tools/__init__.py	Build tool
tools/build_docs.py	Build tool
tools/build_links.py	Build tool
tools/build_notebook.py	Build tool
tools/build_reports.py	Build tool
tools/build_submission_pdfs.py	Build tool
tools/organize_repo.py	Build tool
tools/report_content.py	Build tool
tools/sync_film.py	Build tool
tools/sync_portal.py	Build tool
tools/sync_submission.py	Build tool

THE DATA, AS ISSUED (7)

data/raw/construction_unit_rates_aed.csv	Source dataset
data/raw/dubai_climate_normals_ncm.csv	Source dataset
data/raw/dubai_population_dsc_2023.csv	Source dataset
data/raw/site_zoning_schedule.csv	The measured room schedule
data/raw/sources.json	Every source dataset, with its period
data/raw/species_water_carbon_rates.csv	Source dataset
data/raw/walk_catchment_rings.csv	Source dataset

THE DATA, AS PROCESSED (6)

data/processed/cost_plan.csv	The capital cost plan, line by line
data/processed/hourly_climate_comfort_8760.csv	The 8,760-hour climate and comfort series
data/processed/masterplan_geometry.json	The plan geometry, as exported
data/processed/planting_layout.csv	Every one of the 131 trees
data/processed/schema.json	Generated by the pipeline
data/processed/spatial_grid_comfort.csv	The 15,000-cell ground grid

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded

THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Al Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD_PARKS_MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (1)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.